

Lab Assignment 2: Introduction to Modern Symmetric Ciphers

Introduction

This lab assignment introduces you to key components of modern block ciphers. You will implement routines for basic cryptographic operations such as splitting and combining words, bit manipulation, and substitution-permutation operations. Use Python, Java, or C for the implementation.

Assignment Tasks

Task 1: Splitting an n -bit Word

Write a program to split an n -bit word into two $n/2$ -bit words.

Example:

- Input: 240 (8 bits: 11110000)
- Output: Left Half = 15 (4 bits: 1111), Right Half = 0 (4 bits: 0000)

Task 2: Combining Two $n/2$ -bit Words

Write a program to combine two $n/2$ -bit words into a single n -bit word.

Example:

- Input: Left Half = 15 (4 bits: 1111), Right Half = 0 (4 bits: 0000)
- Output: Combined Word = 240 (8 bits: 11110000)

Task 3: Swapping Left and Right Halves

Write a program to swap the left and right halves of an n -bit word.

Example:

- Input: 240 (8 bits: 11110000)
- Output: 15 (8 bits: 00001111)

Task 4: Circular Shift of an n -bit Word

Write a program to circularly shift an n -bit word by k bits to the left or right.

Example:

- Input: Word = 12 (4 bits: 1100), Shift = 2, Direction = Left
- Output: 3 (4 bits: 0011)

Task 5: Permutation Using a P-box

Define a permutation table for an 8-bit word and write a program to permute the bits based on the table.

Example:

- Input: Word = 202 (8 bits: 11001010), Table = [5, 1, 7, 4, 8, 2, 6, 4]
- Output: 165 (8 bits: 11100100)

Task 6: Inverting a Permutation Table

Given a permutation table, compute its inverse by determining where each bit originated.

Example:

- Input: [6, 3, 4, 5, 2, 1]
- Expected Output: [6, 5, 2, 3, 4, 1]

Steps to Compute the Inverse Table:

1. Label each position in the given table.
2. Swap indices with values.
3. Sort based on indices to get the inverse permutation.

Figure 5.6 *Inverting a permutation table*

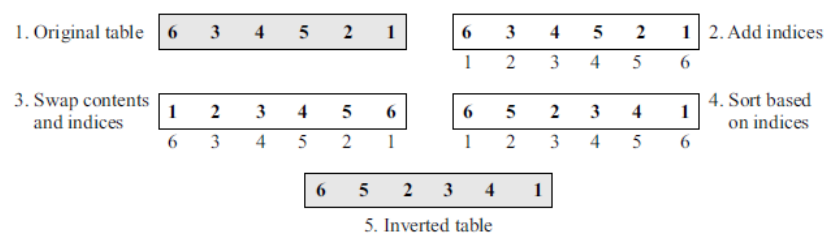


Figure 1: Inverting a permutation table

Implementation: Write a program in Python, Java, or C that:

- Accepts a permutation table as input.
- Computes the inverse permutation.
- Outputs the inverted table.

Submission Instructions

1. Submit your source code files with comments explaining the logic on GitHub .
2. Include test cases for all routines and their outputs.
3. Prepare a short report summarizing your implementation and results with Screenshots.

Grading Criteria

- Correctness of implementation: 60%
- Code readability and comments: 10%
- Test cases and output validation: 10%
- Report quality: 20%